DECISION TREE (REGRESSION)

In [1]:
```
import pandas as pd
df=pd.read_csv(r"C:\Users\Admin\Desktop\student_performance.csv")
df
```

Out[1]:

	weekly_self_study_hours	attendance_percentage	class_participation	total_marks	r
0	18.5	95.6	3.8	97.9	
1	14.0	80.0	2.5	83.9	
2	19.5	86.3	5.3	100.0	
3	25.7	70.2	7.0	100.0	
4	13.4	81.9	6.9	92.0	
...	...	...	...	...	
999995	18.0	95.5	4.8	100.0	
999996	15.7	82.7	6.7	88.3	
999997	14.2	85.1	5.8	87.9	
999998	25.3	90.0	5.3	100.0	
999999	18.3	84.6	4.1	100.0	

1000000 rows × 5 columns

In [2]:
```
# UNIQUE VALUES IN RESULT
df['result'].unique()
```

Out[2]:
```
array(['Pass', 'Fail'], dtype=object)
```

In [3]:
```
# TRANSFORM CATEGORICAL DATA INTO NUMERIC
from sklearn.preprocessing import LabelEncoder
l=LabelEncoder()
```

In [4]:
```
df['result']=l.fit_transform(df['result'])
```

In [5]:
```
# DATA TRANSFORMED (PASS=1, FAIL=0)
df['result'].unique()
```

Out[5]:
```
array([1, 0])
```

In [6]:
```
# DECISION TREE (REGRESSION)
# STEP 1: ASSIGNING VARIABLES(INDEPENDENT/DEPENDENT)
```

```
# INDEPENDENT VARIABLE
x=df[['weekly_self_study_hours','attendance_percentage','class_participation','tota

# DEPENDENT VARIABLE
y=df[['result']]
```

In [7]:
```
# STEP 2:IMPORT TRAIN TEST SPLIT
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)
```

In [8]:
```
print("total data: ",df.shape)
print("x_train data: ",x_train.shape)
print("x_test data: ",x_test.shape)
print("y_train data: ",y_train.shape)
print("y_test data: ",y_test.shape)
```

```
total data:  (1000000, 5)
x_train data:  (700000, 4)
x_test data:  (300000, 4)
y_train data:  (700000, 1)
y_test data:  (300000, 1)
```

In [9]:
```
# STEP 3: APPLY DECISION TREE (REGRESSION)
from sklearn.tree import DecisionTreeRegressor
regressor=DecisionTreeRegressor()
regressor
```

Out[9]:
```
DecisionTreeRegressor
▶ Parameters
```

In [10]:
```
# STEP 4 :FIT TO DATA
# Use only train(x_train,y_train)
regressor.fit(x_train,y_train)
```

Out[10]:
```
DecisionTreeRegressor
▶ Parameters
```

In [11]:
```
# CHECK ACCURACY
regressor.score(x_test,y_test)
```

Out[11]:
```
0.9957251657768529
```

In [12]:
```
# PREDICTION OF NEW DATA
# IMPORT NUMPY FIRST(DATA MUST IN ARRAY FORM)
import numpy as np
data=np.array([[11.7,100.0,8.2,69.8]])
data
```

Out[12]:
```
array([[ 11.7, 100. ,   8.2,  69.8]])
```

In [13]:
```
prediction=regressor.predict(data)
prediction

if prediction == 0:
    print("Fail")
elif prediction ==1:
    print("Pass")
else :
    print("Error")
```

```
Fail
C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but DecisionTreeRegressor was fitted with feature names
  warnings.warn(
```

In [ ]: